# Open problems:
# The Born rule is equivalent to the von Neumann entropy

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

No item below blocks the verification of a main claim of the brief.

## From the brief

1. **Infinite entropy.** EQST takes the entropy to be real-valued, but in infinite dimension the von Neumann entropy of some density operators, and the Shannon entropy of some countable families of weights, is $+\infty$. With real values BR-ENT and [B]MUTEX cannot hold in infinite dimension. Which codomain is intended, and whether [B]MUTEX should be restricted to ensembles of finite entropy, is open. The formalization uses $[0, \infty]$. (Brief, Sec. 3, EQST; Log, "Problems found", first item.)